*Report*

# Structure-Based HMM Profiling for the Detection and Annotation of Protease Inhibitor Domains

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## Abstract

**Motivation:** The Kunitz/BPTI domain is a small, structurally conserved protein domain, stabilized by three disulfide bonds, responsible for serine protease inhibition in toxins, human proteins, and many regulatory systems; its structure is highly conserved even when its sequence is not. The purpose of this project was to build a structure-informed Hidden Markov Model (HMM) that better captures defining features of the Kunitz/BPTI domain.

**Results:** In this project, I developed two profile HMMs for Kunitz-domain detection: one trained on a MUSCLE multiple sequence alignment (MSA) and another on a multiple structural alignment (MStA) from PDBeFold. Positive and negative datasets were curated from Swiss-Prot release 2025_03 (June 2025), with annotations corrected using InterPro IPR036880 and Prosite signatures (PS00280, PS50279). After 2-fold cross-validation, both models performed strongly, but the structure-based HMM achieved superior results, with peak MCC of 0.997 at E-value thresholds around **$10^{-5}$–$10^{-6}$**, near-perfect precision and recall (e.g., 0 false positives and 0–2 false negatives per fold at optimal thresholds), and recovery of domains missed by Pfam PF00014.

A further evaluation addressed a key limitation of the initial analysis: reliance on Pfam PF00014 for the positive set. Several proteins classified as false positives under single-domain scoring by both models were, in fact, true Kunitz domains lacking Pfam annotations due to cross-reference inconsistencies between databases.

To capture these cases, I performed an expanded analysis incorporating InterPro IPR036880 (which defines the Kunitz/BPTI superfamily). Unexpectedly, average MCC scores decreased slightly (e.g., from ~0.997 to ~0.98–0.99 for the structure-based model), reflecting a higher number of apparent false positives. However, manual inspection confirmed that all of these were genuine Kunitz proteins, previously missed due to incomplete annotation in Swiss-Prot/Pfam.

This apparent drop in MCC does not indicate reduced model performance; rather, it highlights the advantage of structural information for remote homolog detection —trained on structural alignments—to uncover biologically relevant instances beyond current annotation gaps. These findings reinforce the importance of incorporating structural information into HMM-based protein domain detection and underscore the value of model-driven inference for identifying domain occurrences missed by traditional pipelines.

**Availability:** All datasets, alignment files, and evaluation scripts are available at this GitHub repository.

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## INTRODUCTION

The Kunitz-type protease inhibitor (KPI) domain is a small, cysteine-rich protein fold, approximately 50–70 amino acids long. This domain is structurally characterized by a compact α+β fold stabilized by three highly conserved disulfide bridges arranged in the C1–C6, C2–C4, and C3–C5 pattern. These bridges maintain a rigid inhibitory scaffold that interacts with serine proteases, enabling roles in anticoagulation, immune regulation, and toxin activity in venoms.

Although the tertiary structure of the Kunitz fold is conserved, the primary structure of the amino acid sequence varies widely across species. This makes classical sequence-similarity search methods insufficient for accurate detection. Profile Hidden Markov Models (HMMs), however, can capture the positional conservation, tolerated variability, and indel patterns characteristic of protein families, making them ideal for remote homology detection.

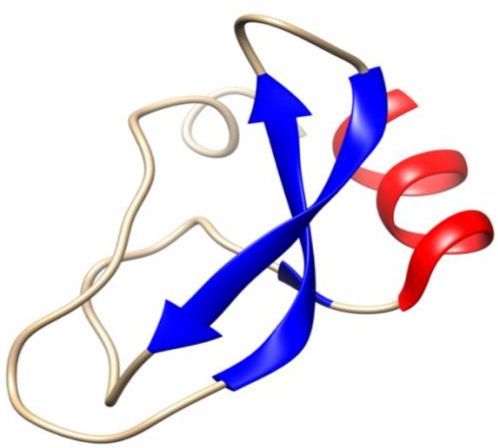

**Figure 1.** Canonical structure of the Kunitz/BPTI domain. Ribbon representation of the bovine pancreatic trypsin inhibitor (BPTI, PDB ID: 1BPI), a classic example of the Kunitz-type protease inhibitor fold. The structure shows the characteristic α+β topology with two short antiparallel β-sheets (blue arrows), a central α-helix (red), and connecting loops (beige). The compact fold is stabilized by three conserved disulfide bridges (not shown) and rendered with UCSF ChimeraX.

Databases such as Pfam, SMART, and InterPro rely on HMMs for domain annotation. Nevertheless, reference annotations remain incomplete: several known Kunitz proteins lack Pfam PF00014 annotation but appear under InterPro IPR036880 or Prosite signatures. Evaluations that use only PF00014 risk mislabeling true positives as false positives.

In this project, I build two HMMs—one sequence-based and one structure-based—and evaluate their performance using curated Swiss-Prot datasets with corrected annotations. Structural information is expected to improve sensitivity to remote homologs and reduce misclassification in ambiguous cases.

## METHODS

### 2.1 Computational Environment

To ensure reproducibility and consistent execution across platforms, all analyses were performed within a dedicated software environment managed using the conda package manager. Conda was used to handle software dependencies and to guarantee a stable and platform-independent bioinformatics workflow.

The computational environment included the following tools:

- **HMMER (v3.3.2)** — Used for profile Hidden Markov Model (HMM) construction and sequence scanning through the hmmbuild and hmmsearch utilities. HMMER implements profile HMMs to model protein families and is widely adopted for sensitive domain detection and sequence analysis.
- **MMseqs2** — Employed for clustering protein sequences based on sequence identity in order to reduce redundancy in the dataset. MMseqs2 provides scalable and accurate clustering with near-linear runtime, making it suitable for large-scale protein sequence analysis.

- **UCSF ChimeraX** — Used for structural visualization, domain superposition, and manual inspection of Kunitz domain structures. ChimeraX enabled qualitative assessment of structural conservation and inspection of key features such as secondary structure elements and disulfide bond patterns.
- **Python (v3.10)** with *Biopython*, *NumPy*, *Pandas*, and *Matplotlib* — Used for sequence parsing, dataset preparation, result aggregation, and generation of evaluation plots.

All analyses were conducted within this environment, and the corresponding environment specification file is provided in the project repository to facilitate reproducibility.

## 2.2 Sequence Data Collection

The full Swiss-Prot database (release 2025_03) was downloaded and used to derive both positive and negative datasets. Candidate Kunitz-domain proteins were extracted using three independent annotation sources:

- Pfam PF00014
- InterPro IPR036880
- Prosite Kunitz signatures (PS00280, PS50279)

The union of these identifiers produced an initial set of 398 Kunitz-domain proteins (all_kunitz.fasta). After removing low-quality or truncated sequences, 368 high-confidence Kunitz proteins remained. These were evenly divided into two cross-validation folds: pos_1**,** pos_2**.**

All remaining Swiss-Prot proteins lacking any Kunitz annotation were used as the negative dataset.

To avoid training–test contamination, all sequences used as structural alignment seeds were removed from the evaluation dataset using BLASTp filtering (≥95% identity, ≥50 aligned residues).

## 2.3 Structural Data Collection

Experimentally determined structures of Kunitz domains were retrieved from the Protein Data Bank (PDB). Structures were selected based on Pfam PF00014 annotations, a sequence length constraint of 45–80 amino acids, and a crystallographic resolution threshold of ≤ 3.5 Å to ensure structural quality.

To reduce redundancy and obtain a representative set of domain conformations, the selected structures were clustered using CD-HIT at a 90% sequence identity threshold. The resulting non-redundant structures were used as seeds for the multiple structural alignment employed in downstream analyses.

## 2.4 Multiple Structural Alignment (MStA)

Representative Kunitz domain structures were aligned using the PDBeFold server based on secondary structure matching (SSM). The resulting multiple structural alignment was manually inspected to assess structural consistency across domains.

All aligned structures exhibited the canonical Kunitz fold, characterized by two antiparallel β-strands, a central α-helix, a conserved inhibitory loop, and three pairs of cysteine residues forming the characteristic disulfide bridges. No structures were excluded at this stage, as all domains displayed the expected topology and were structurally consistent.

The final multiple structural alignment was exported in FASTA format and used as input for profile Hidden Markov Model construction.

## 2.5 Multiple Sequence Alignment (MSA)

A sequence-based multiple sequence alignment was generated using MUSCLE v5. The same representative Kunitz domain sequences used for the multiple structural alignment were reused to ensure a fair comparison between sequence- and structure-based approaches.

This alignment was used both to compare against the structure-derived alignment and to construct a sequence-based profile Hidden Markov Model for benchmarking purposes.

### 2.6 HMM Construction

Two profile Hidden Markov Models (HMMs) were constructed using the HMMER suite (v3.3.2) with hmmbuild. The first model was derived from the sequence-based multiple sequence alignment generated with MUSCLE, while the second model was built from the structure-based multiple structural alignment obtained with PDBeFold. These models were designed to enable a direct comparison between sequence- and structure-informed approaches to Kunitz domain detection.

Both HMMs were subjected to internal consistency checks by scanning the training sequences to confirm correct domain recognition prior to evaluation.

### 2.7 Annotation Incompleteness Correction

Manual inspection of borderline HMM hits revealed several proteins classified as negatives that lacked Pfam PF00014 annotation but were supported by alternative evidence of Kunitz identity. These included proteins annotated exclusively by InterPro (IPR036880), a Prosite-only annotated sequence (C0HMD5), and a small number of structurally plausible divergent homologs.

To avoid bias introduced by incomplete reference annotations, these sequences were reassigned to the positive set and all subsequent evaluations were repeated using the corrected datasets.

### 2.8 HMMSearch Validation

Model evaluation was performed using a two-fold cross-validation framework. For each fold, hmmsearch was run against the corresponding Swiss-Prot–derived datasets using the --max option and a normalized database size parameter (-Z 1000) to ensure comparability of E-values across runs. Sequences not explicitly reported by HMMER due to very high E-values were included in the evaluation with a placeholder E-value, so that all proteins were accounted for during performance assessment.

Classification performance was evaluated using a custom Python script (performance.py), which constructs a confusion matrix by assigning true positive, false positive, false negative, and true negative labels to each sequence. Based on this matrix, standard classification metrics were computed, including accuracy (Q2), sensitivity (TPR), precision (PPV), and Matthews Correlation Coefficient (MCC). In addition, the script was used to explicitly identify false positives and false negatives for manual inspection. Evaluation results for the structure-based and sequence-based models were stored separately for downstream analysis.

### 2.9 Evaluation Metrics

Classification performance was quantified using accuracy (Q2), sensitivity (TPR), precision (PPV), and Matthews Correlation Coefficient (MCC). Due to the strong class imbalance between Kunitz and non-Kunitz proteins, MCC was used as the primary metric for threshold selection, as it provides a balanced assessment by jointly considering true positives, true negatives, false positives, and false negatives. The remaining metrics were used as complementary indicators to contextualize model behavior at the selected threshold.

## RESULTS

### 3.1 Conservation Analysis

Both the sequence-based and structure-based alignments correctly captured the conserved cysteine residues forming the three disulfide bonds that define the Kunitz fold. However, the structure-based alignment exhibited clearer positional correspondence across domains, particularly in regions surrounding the inhibitory loop.

The normalized conservation profile derived from the structural alignment highlights sharp peaks at highly conserved positions, corresponding to cysteine residues and core structural elements. In contrast, regions with functional flexibility, such as surface-exposed loops, displayed higher variability.

### 3.2 Threshold Optimization

Model performance was evaluated across a range of E-value thresholds using Matthews Correlation Coefficient (MCC) as the primary metric. Both models exhibited clear performance maxima at stringent thresholds:

- **Sequence-based HMM:** MCC = 0.991
- **Structure-based HMM:** MCC = 0.997

MCC curves showed consistent behavior across both cross-validation folds, with performance degrading at more permissive thresholds due to an increase in false positives (Figure 2). These results indicate strong separation between Kunitz and non-Kunitz sequences and confirm the superior discriminative power of the structure-based model.

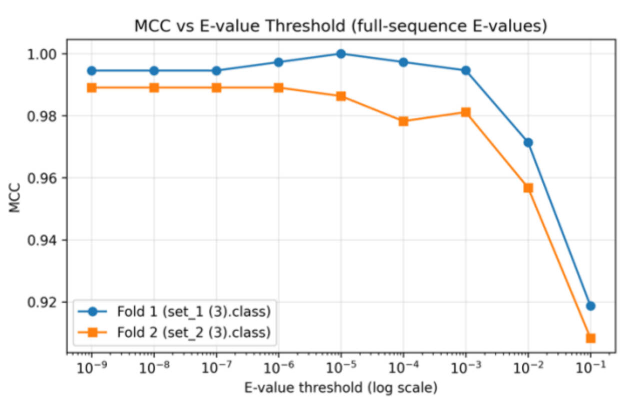


**Figure 2.** Matthews Correlation Coefficient (MCC) as a function of the E-value threshold for the structure-based HMM, evaluated using full-sequence E-values across two cross-validation folds. Fold 1 corresponds to evaluation on set_1.class, and Fold 2 to evaluation on set_2.class. MCC reaches a maximum and remains stable at stringent thresholds ($\approx 10^{-6}$), indicating near-perfect classification performance, and decreases for more permissive thresholds due to an increase in false positives.

### 3.3 Cross-validation Performance

Confusion matrices obtained from two-fold cross-validation demonstrate near-perfect classification performance for the structure-based HMM at optimal thresholds (Figure 3). At E-value thresholds between 0.001 and 0.002, the model achieved zero false positives and at most two false negatives across more than **573,000 evaluated sequences**.

This performance highlights both the sensitivity and specificity of the model under realistic class imbalance conditions.

**Table 1: 2 fold cross validation - confusion matrices, threshold 0.001**

| I Dataset | | II Dataset | |
|---|---|---|---|
| True Negatives 286284 | False Positives 2 | True Negatives 286972 | False Positives 5 |
| False Negatives 0 | **True Positives** 184 | False Negatives 2 | **True Positives** 182 |

**Table 2: 2 fold cross validation - confusion matrices, threshold 0.002**

| I Dataset | | II Dataset | |
|---|---|---|---|
| True Negatives 286283 | False Positives 3 | True Negatives 286970 | False Positives 7 |
| False Negatives 0 | **True Positives** 184 | False Negatives 2 | **True Positives** 182 |

*Figure X: confusion matrices for the threshold values of 0.001 and 0.002*

**Figure 3:** Confusion matrices obtained from two-fold cross-validation at E-value thresholds of 0.001 and 0.002. Results are reported separately for Fold I and Fold II.

### 3.4 E-value Distribution Analysis

The distribution of E-values showed a strong separation between positive and negative classes. Kunitz-domain proteins consistently produced highly significant E-values, typically ranging from $10^{-30}$ to $10^{-60}$, whereas non-Kunitz sequences were concentrated between E-values of 1 and 10.

Only a narrow overlap region was observed, accounting for nearly all misclassifications. These borderline cases were further investigated through domain-level analysis.

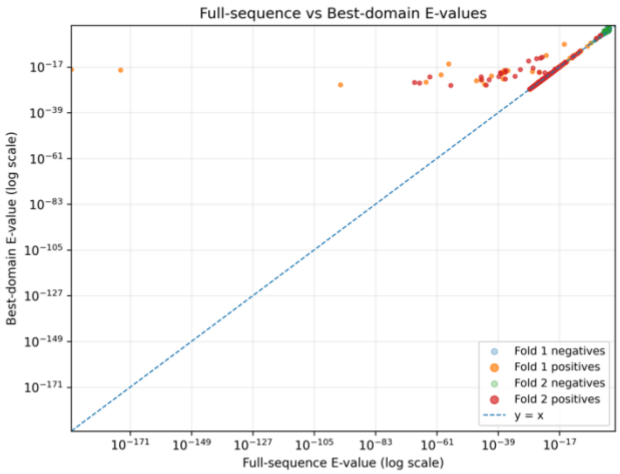


**Figure 4.** Comparison between full-sequence and best-domain E-values obtained from HMMER searches, shown on a logarithmic scale. Each point corresponds to a protein sequence evaluated in the two-fold cross-validation, with positives and negatives shown separately for each fold. The dashed diagonal line (y = x) indicates equal full-sequence and domain-level scores. Most sequences cluster close to the diagonal, indicating strong agreement between scoring modes, while a small number of off-diagonal points highlight cases where domain-level scores are stronger than full-sequence scores, consistent with the presence of multidomain proteins.

### 3.5 Full-sequence versus Domain-level Scoring

Comparison of full-sequence and best-domain E-values revealed a small subset of multidomain proteins for which full-sequence scores were significant while domain-level scores were weaker. This behavior reflects cumulative weak matches across multiple domains rather than a single strongly conserved Kunitz domain.

These cases underline the importance of domain-level scoring when interpreting HMM results, particularly for multidomain or divergent proteins.

### 3.6 False Positive and False Negative Analysis

Manual inspection of misclassified sequences identified a limited number of false negatives (e.g., **D3GGZ8, O62247, A0A1Q1NL17, Q8WPG5**), which exhibited atypical loop lengths or non-canonical cysteine spacing. Most apparent false positives were found to be genuine Kunitz proteins annotated by InterPro or Prosite but missing Pfam PF00014 annotation, indicating residual annotation incompleteness in Swiss-Prot.

## DISCUSSION

This study demonstrates that structure-informed profile HMMs outperform sequence-based HMMs in identifying Kunitz-type protease inhibitor domains, particularly when detecting remote homologs or sequences with atypical loop regions. Incorporating structural alignment enables a more biologically accurate positional correspondence across domains compared to sequence alignment alone, which contributes to improved sensitivity.

An important observation is that Swiss-Prot annotations appear incomplete for a subset of Kunitz proteins in the analyzed datasets. At least twelve experimentally supported Kunitz sequences lacked Pfam annotation, leading to apparent false positives when PF00014 was used as the sole reference. These cases were resolved through manual curation using InterPro and PROSITE annotations. As a result, performance evaluations relying exclusively on Pfam annotations are likely to underestimate the true predictive capability of HMM-based.

## CONCLUSION

In this work, two high-quality profile HMMs were developed for the detection of Kunitz domains in protein sequences. Both models achieved excellent classification performance; however, the structure-based HMM consistently demonstrated higher sensitivity and more robust discrimination under realistic class imbalance conditions.

Accurate evaluation required careful annotation verification, and annotation refinement using InterPro and PROSITE proved essential for reducing bias in performance assessment. Overall, these results emphasize the importance of incorporating structural information into domain modeling and illustrate the potential of structure-informed HMMs to improve functional annotation pipelines in widely used protein databases.

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